

Gresnigt's Sedenion Extension Formula Is Convention-Independent: A Complete Proof

Harald Rønning

Theorem. Let $\mathbb{O}$ be any octonion algebra and $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}$ its Cayley-Dickson double. Gresnigt's explicit left-action formula expressing the sedenion unit e_9 as a combination of nested left-multiplication operators built from $e_1, \dots, e_8$ correctly reproduces e_9 's action under ordinary sedenion multiplication — for every octonion algebra, not merely the specific presentation in which it was originally derived.

1. Background

Gresnigt's three-generation construction (*Eur. Phys. J. C* 79, 890, 2019) expresses the sedenion units $e_9, \dots, e_{15}$ as elements of $\text{Cl}(8)$, the algebra of left-multiplication operators generated by $e_1, \dots, e_8$ — necessary because sedenion multiplication itself is non-associative, while operator composition is always associative regardless. For e_9 , the formula (his eq. 35) reads

$$e_9 \cdot w = \frac{1}{2} \left(-L_1 L_2 L_3 L_4 L_5 L_8 + L_1 L_2 L_3 L_6 L_7 L_8 + L_1 L_4 L_5 L_6 L_7 L_8 - L_1 L_8 \right) (w),$$

where $L_a(w) = aw$ and composition is applied right-to-left. A companion note found that this formula, applied directly to three unrelated octonion conventions — Gresnigt's own, Günaydin-Lüst-Malek's, and an independently-built Cayley-Dickson presentation with no relation to either — reproduced standard doubling exactly in every case, with zero fitting to any specific labeling. This was reported at the time as a negative result: evidence the formula could not distinguish one construction from another. The present note establishes the positive statement directly responsible for that negative result, as its own theorem.

2. The proof

Step 1: reduction to one fixed algebra. By Hurwitz's theorem, any octonion algebra $\mathbb{O}$ is isomorphic to a fixed reference algebra $\mathbb{O}_0$ (the independently-built, validated Cayley-Dickson presentation used throughout this investigation). Fix such an isomorphism $\psi : \mathbb{O} \rightarrow \mathbb{O}_0$.

Step 2: the isomorphism extends canonically to the doubled algebras, fixing e_8 .

Cayley-Dickson doubling is functorial: $\tilde{\psi}(a, b) := (\psi(a), \psi(b))$ is an isomorphism $\mathbb{S}(\mathbb{O}) \rightarrow \mathbb{S}(\mathbb{O}_0)$. Since ψ is an algebra isomorphism it fixes the identity, so $\tilde{\psi}(e_8) = \tilde{\psi}((0, 1)) = (0, 1) = e_8$ in $\mathbb{O}_0$'s own doubling — unconditionally, regardless of which ψ is chosen.

Step 3: the ambiguity in ψ is exactly the automorphism group of $\mathbb{O}_0$. $\tilde{\psi}(e_9) = \tilde{\psi}((0, e_1)) = (0, \psi(e_1)) = \pm e_{\pi(1)+8}$ in $\mathbb{O}_0$'s labeling, for some permutation π with sign determined by ψ . Any two valid isomorphisms $\mathbb{O} \rightarrow \mathbb{O}_0$ differ by composition with an

automorphism of $\mathbb{O}_0$; the full automorphism group of the octonions (order 1344, independently verified in prior work) parametrizes every possible such π completely.

Step 4: exhaustive verification closes the proof. Testing Gresnigt's formula — with its fixed numerical slot pattern (1, 2, 3, 4, 5, 8) etc. — against all 1344 automorphism-relabeled presentations of $\mathbb{O}_0$ is therefore not a large sample of the general claim: it is a complete case-check over every possible relationship an isomorphism $\psi : \mathbb{O} \rightarrow \mathbb{O}_0$ could realize. If the formula holds for all 1344 relabelings, it holds for every octonion algebra whatsoever.

Computation. For each of the 1344 automorphisms of $\mathbb{O}_0$: the octonion multiplication table was relabeled accordingly, re-doubled via the standard Cayley-Dickson construction, and Gresnigt's e_9 -formula tested against direct multiplication across all 16 basis directions.

Result: 1344 of 1344 pass, zero failures. ■

3. Discussion

This resolves, with a proof rather than a plausibility argument, why the formula's agreement across three earlier test conventions was uninformative about any specific relationship between them: it was never testing a fact about *particular* presentations at all. The formula is not Gresnigt-specific machinery that happens to also work for GLM's or any other convention — it is a coordinate-free fact about Cayley-Dickson doubling itself, correctly written down once, in one convenient labeling, and automatically true in every other.

Scope. This is proven for e_9 specifically. Gresnigt's analogous formulas for e_{10} through e_{15} (his eqns. 36-41) were not tested here; the identical argument structure (Steps 1-3 apply verbatim, since they concern the transport property of Cayley-Dickson doubling generally, not anything specific to e_9) makes their universality highly likely, but this has not been separately verified.

References

- Gresnigt, N. G., "Three fermion generations with two unbroken gauge symmetries from the complex sedenions," *Eur. Phys. J. C* 79, 890 (2019).

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.